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Diversification and Evolutionary Dynamics in Tropical Montane Regions

Jhan C. Salazar[1] | Adam C. Algar[2] | Steven Poe[3] | Jonathan B. Losos[1] | Julián A. Velasco[4]

[1]Department of Biology, Washington University, St. Louis, Missouri, USA | [2]Department of Biology, Lakehead University, Thunder Bay, Ontario, Canada | [3]Department of Biology, University of New Mexico, Albuquerque, New Mexico, USA | [4]Instituto de Ciencias de la Atmósfera y Cambio Climático, Universidad Nacional Autónoma de México, México City, México

Correspondence: Jhan C. Salazar (jhancsalazar@wustl.edu) | Julián A. Velasco (javelasco@atmosfera.unam.mx)

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ABSTRACT

Aim: The evolution of montane species provides critical insights into the drivers of adaptation and diversification. Topographic complexity, a defining feature of many mountainous landscapes, promotes ecological and geographic isolation, often accelerating speciation rates. However, the extent to which topographic complexity directly shapes diversification remains unresolved. Here, we investigate the evolutionary dynamics of *Anolis* lizards across Neotropical mountain ranges to test two hypotheses: (1) higher elevation environments promote higher speciation rates in species that inhabit them; and (2) greater topographic complexity and climatic stability (i.e., lower past climatic-change velocity) positively influence speciation rates.

Location: The Americas.

Time Period: Present.

Major Taxa Studied: *Anolis* lizards.

Methods: We gathered topographic complexity and past climatic-change velocity data for 303 anole species and performed phylogenetic analyses to assess how speciation rates change across mountain ranges in the Americas.

Results: We found that topographic complexity and past climatic-change velocity do not significantly influence speciation rates, diverging from prior studies that link rugged landscapes or past climatic-change velocity to high diversification rates.

Main Conclusions: Our findings challenge assumptions about the direct role of topographic complexity in speciation, highlighting the need to consider multifaceted ecological and biogeographical factors driving evolutionary processes. By disentangling the relative contributions of climatic stability (i.e., lower past climatic-change velocity) and topographic heterogeneity, this study highlights the complex dynamics shaping biodiversity in tropical mountains and calls for further integrative approaches to understand species diversification in the face of climatic and geological change.

1 | Introduction

Mountains cover only a quarter of Earth's land surface (Kapos et al. 2000), yet they harbour a disproportionate share of biodiversity, and many global biodiversity hotspots (Sonne and Rahbek 2024). Aside from their diversity, montane regions possess pronounced environmental gradients (e.g., topographic complexity, climatic stability) that promote diverse microhabitats

Jonathan B. Losos and Julián A. Velasco contributed equally to this work.

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and strong selective pressures, driving ecological divergence and reproductive isolation (Hoorn et al. 2010; Rahbek, Borregaard, Colwell, et al. 2019). These conditions make montane regions ideal for studying diversification.

One of the primary factors often suggested as driving diversification in the region is its pronounced topographic complexity, which has been proposed to generate and maintain biological diversity across spatial and temporal scales (e.g., Amatulli et al. 2018; García-Rodríguez et al. 2025). Topographic complexity could promote geographic and ecological isolation, thereby accelerating speciation (Amatulli et al. 2018). In this context, topographic complexity encompasses multiple interrelated landscape characteristics, including variation in elevation, relief, slope steepness, erosion rates, and associated climate heterogeneity (Badgley et al. 2017; Rahbek, Borregaard, Colwell, et al. 2019; Antonelli et al. 2018). Together with climatic variation in precipitation and temperature, topographic complexity creates heterogeneous landscapes that facilitate geographic isolation and ecological opportunity (Antonelli et al. 2018; Rahbek, Borregaard, Antonelli, et al. 2019; Sanín et al. 2024). For instance, García-Rodríguez et al. (2025) found a strong positive relationship between speciation rate and climatic complexity in amphibians, suggesting that geographic barriers limit gene flow and promote divergence. However, despite these insights, this is the only study to our knowledge that has explicitly tested how topographic and climatic complexity jointly influence speciation rates across phylogenetic scales in a widespread taxon, leaving the role of environmental heterogeneity in diversification particularly understudied in montane systems.

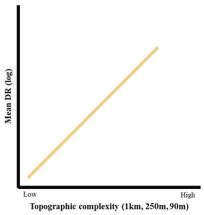
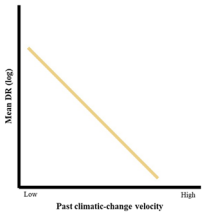
Additionally, climatic stability also influences species diversification by promoting long-term isolation and reducing extinction rates (Dynesius and Jansson 2000; Rolland et al. 2014; Fine 2015; Buckley et al. 2024). In montane regions, the interaction of climatic stability and topographic complexity enhances isolation, supports niche specialisation, and facilitates both allopatric and parapatric speciation through the preservation of microhabitats and ecological gradients (Rahbek, Borregaard, Colwell, et al. 2019; Rahbek, Borregaard, Antonelli, et al. 2019). These dynamics have shaped the unique evolutionary histories of many montane taxa, including amphibians, birds, and reptiles, where narrow-ranged species often persist in stable environments such as cloud forests near tropical coasts (Cadena et al. 2011; Fjeldså et al. 2012; McGuire et al. 2014; Hutter et al. 2013). However, climatic instability may also increase speciation, as observed in mammals, by creating novel ecological opportunities and geographic barriers that drive population divergence (Morales-Barbero et al. 2021). Beyond montane zones, mountain-driven climatic stability has been linked to diversification at regional and global scales.

One of the best examples is the Andes, the longest continental mountain range in the world which has shaped Neotropical biodiversity through high endemism and adaptive radiation across multiple lineages (e.g., Fjeldså et al. 2012; Hazzi et al. 2018; Everson et al. 2020; Scherz et al. 2023). The uplift of the Andes facilitated niche evolution in plants (Dellinger et al. 2024), while elevational diversity gradients help explain global bird diversification (Quintero and Jetz 2018). Moreover, past climatic-change velocity has been shown to influence speciation rates (García-Rodríguez et al. 2025). Together, these findings suggest that long-term climatic dynamics play a fundamental role in shaping biodiversity. Yet, their interactive effects with geology remain less well understood, particularly in mountains where topographic complexity determines whether isolated habitats persist long enough for speciation and lineage survival (Wiens 2004; Pyron et al. 2015). For instance, phylogenetic niche conservatism may drive allopatric speciation when lineages retain ancestral niches but become isolated by elevational barriers created through climate change and topographic uplift (Wiens 2004). Stable climates in rugged landscapes can reinforce such isolation, whereas unstable climates may erode it, emphasising the need to examine climatic stability and topographic complexity together (Pyron et al. 2015; Kozak and Wiens 2010). Our study addresses this gap by investigating how these processes interact at regional scales and over evolutionary timescales (Willis and Whittaker 2002; Pearson and Dawson 2003), offering new insights into how spatial and temporal environmental heterogeneity shape diversification.

Anoles, members of the lizard genus *Anolis*, represent one of the most well-known examples of adaptive radiation in vertebrates, having diversified into more than 400 species distributed across the Neotropics, from lowland forests to high-elevation habitats up to 3700 m above sea level (Ruibal 1961; Rand 1964; Poe et al. 2017; Moreno-Arias et al. 2021). Their evolutionary success has made them a model system in studies of morphological diversification, ecological specialisation, and physiological adaptation (Williams 1972; Pinto et al. 2008; Losos and Ricklefs 2009; Mahler et al. 2010; Muñoz et al. 2014; Conover et al. 2015; Méndez-Galeano and Calderón-Espinosa 2017; Salazar et al. 2019; Muñoz et al. 2023; Salazar et al. 2025). Much of what we know about anole evolution comes from island systems, where repeated evolution of habitat specialists (i.e., "ecomorphs") has shaped community structure through morphological convergence and thermal niche partitioning (e.g., Losos et al. 1998; Mahler et al. 2010; Gunderson et al. 2018). However, anoles are also highly diverse in mainland regions from Mexico to Brazil, including the tropical Andes, where complex topography and steep environmental gradients provide rich opportunities to study evolutionary dynamics in continental montane systems (Velasco et al. 2016, 2018; Salazar et al. 2019; Alencar et al. 2024; Salazar et al. 2025). Anoles vary widely in both their morphology and physiology, often exhibiting specialised traits that reflect adaptation to specific microhabitats and climates (Hertz et al. 2013; Muñoz et al. 2014; Méndez-Galeano and Calderón-Espinosa 2017; Salazar et al. 2019; Bodensteiner et al. 2024; Salazar et al. 2025; Velasco et al. 2026; Moreno-Arias and Méndez-Galeano 2026). Despite extensive research on thermal biology and morphological evolution, key knowledge gaps remain, especially regarding climatic niche specialisation in both island and mainland montane species and the role of topography and climate on diversification. We use this group of lizards to test hypotheses about how geological processes influence speciation rates across montane regions. Given their ecological breadth, repeated radiation, and capacity for local adaptation, *Anolis* lizards provide an ideal system to investigate how montane environments shape recurrent patterns of speciation in tropical ectotherms.

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TABLE 1 | Overview of tested predictors of mountain speciation rates. For each predictor, we provide the hypothesized mechanism, a specific prediction, and a visual depiction of the expected correlation. Modified from García-Rodríguez et al. (2025).

Predictor	Mechanism	References	Predictions	Expected correlations
Topographic complexity	Limited gene flow caused by isolation and climate variability in mountain regions can drive higher speciation rates	García-Rodríguez et al. (2021); Cerezer et al. (2022); García-Rodríguez et al. (2025)	Topographic complexity in tropical mountains is expected to limit gene flow across populations, promoting higher rates of speciation	
Past climatic-change velocity	In climatically stable environments, species have enough evolutionary time for divergence and speciation	Igea and Tanentzap (2021); García-Rodríguez et al. (2025)	Low climatic-change velocity in the tropics may have facilitated longer periods of evolutionary persistence and divergence, resulting in higher speciation	

In this study, we test two primary hypotheses about the role of high-elevation habitats in shaping biodiversity in tropical mountains. First, we hypothesize that high-elevation habitats act as "species pumps" where the interaction between high topographic complexity and low climatic-change velocity increases diversification. Specifically, we expect that montane lineages exhibit higher speciation rates because complex terrain provides "topoclimatic buffering," allowing populations to persist in stable micro-refugia during periods of regional climatic instability. As populations become isolated, restricted gene flow facilitates the divergence necessary for speciation. As a result, we expect that high-elevation species will show higher diversification rates compared to their lowland counterparts. Second, we hypothesize that the frequency of transitions from lowlands to highlands is moderated by paleoclimatic stability. In regions with high climatic stability, the formation of persistent thermal gradients allows for gradual niche evolution. We predict that transition rates into montane zones will be highest where topographic complexity is coupled with low past climatic velocity, as these conditions minimise the risk of extinction during the adaptive shift to colder, high-elevation environments (Table 1).

2 | Methodology

2.1 | Phylogenetic and Georeferenced Data

Phylogenetic analyses were conducted using the latest comprehensive time-calibrated *Anolis* phylogeny (Poe et al. 2017). Our analyses included 303 species and were based on molecular data from Poe et al.'s Maximum Clade Credibility (MCC) tree of 322 species.

Currently, there is no consensus definition for what constitutes a highland species (but see Linck 2025). For this reason, we employed three different elevational thresholds to distinguish between lowland and highland species: 300, 700 and 1500 m. We chose 700 m (hereafter "700 m criterion") as our primary criterion because this point has been associated with significant elevational turnover among Caribbean anole species (Frishkoff et al. 2022) but included the other two in Data S1 to test the sensitivity of our results to the particular criterion chose. The 300 m threshold (hereafter "300 m" criterion) was chosen because it aligns with geomorphological definitions of mountainous terrain; most geologists classify a mountain as a landform that rises at least 300 m or more above its surrounding area (Körner et al. 2016). The 1500 m threshold (hereafter 1500 m criterion) captures transitions commonly observed in vegetation and species composition around 1500–1650 m (e.g., Hu et al. 2018; Irungbam et al. 2025). These classifications allow us to explore how climatic stability and topographic complexity interact across elevational zones to shape diversification.

The locality data for *Anolis* species were obtained from Velasco et al. (2020). These data are publicly available at https://doi.org/10.6084/m9.figshare.11778399.v1. Additionally, our analysis included several updates, corrections, or the addition of new occurrence data for a few species; in total, there were 361 species, but only 303 were present in Poe et al.'s (2017) phylogenetic tree.

2.2 | Topographic Complexity and Past Climatic-Change Velocity

For each of the 24,360 individuals in our occurrence dataset, we used the roughness metric from Amatulli et al. (2018) as a proxy for topographic complexity. This metric serves as a descriptor of terrain configuration and surface physical complexity. Using the *extract()* function in the *terra* package for R (Hijmans 2025), we obtained roughness values from the Amatulli et al. (2018) dataset at three resolutions: 1 km, 250 m, and 90 m. Finally, these individual values were averaged by species to calculate a species-level mean roughness.

García-Rodríguez et al. (2025) calculated historical climate-change velocity using past climate data across 11-time horizons from the Paleoclim database (Brown et al. 2018), which covers 11 time periods defined by abrupt climate events over the last

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approx. 3 million years (Brown et al. 2018). This metric captures the velocity of local climate shifts (km/year) in both temperature and precipitation variables. Following this, García-Rodríguez et al. (2025) calculated past climate-change velocity for each period relative to current conditions and then averaged these values across all periods. After we extracted the past climatic-change velocity data for each of the 24,360 individuals in our occurrence records data set, we then averaged these values for each species to obtain a mean for the past climatic-change velocity for each species.

2.3 | Macroevolutionary Dynamics and Diversification Shifts

We tested whether the observed signal could be explained by shifts in diversification rates across the phylogeny of extant species, using a reversible jump MCMC (rjMCMC) in CoMET (CPP [compound Poisson process] on mass-extinction times) functions, particularly the *tess.analysis()* function of the R package *TESS* (Höhna et al. 2016; May et al. 2016). We used the anole tree, with empirical hyperpriors on speciation and extinction rates –the argument *empiricalHyperPriors* in the *tess.analysis()* function was set to TRUE–, a prior expectation of one diversification rate shift, and 84% uniform taxon sampling. We started the MCMC using the software's autotuning and autostopping to terminate chains after 500 effective samples, with 2,000,000 iterations, a burn-in of 100,000 and a thinning interval of 4000, repeated the analysis four times, and confirmed convergence and stability across runs. Sensitivity tests varying priors and sampling assumptions showed our results were robust, meaning that our findings are not sensitive to the specific sampling fraction or the prior on the number of rate changes (see Data S1 and Table S1). We also verified that posterior estimates were informed by the data by sampling from the joint prior.

2.4 | Geography-Dependent Diversification (GeoHiSSE)

To test whether high-elevation habitats act as "species pumps," we employed Geographic State Speciation and Extinction (GeoSSE) and Hidden Geographic State Speciation and Extinction (GeoHiSSE) models (Caetano et al. 2018). This framework allows us to estimate parameters for speciation, extinction, and dispersal specific to each elevational zone. We categorized species into three states based on elevation: lowland endemics (state: 00), highland endemics (state: 11), and a widespread state of species that share both lowlands and highlands (state: 01), which represents a free parameter. For this analysis, we treated all highland regions as equivalent, assuming that the shared environmental and topographic characteristics of high-elevation habitats exert a consistent influence on diversification across the study area (Iturralde-Vinent 2006; Antonelli et al. 2018). We acknowledge this assumption and further explore its implications in the Discussion.

We used the GeoHiSSE framework to account for both observed geographic ranges and unobserved (hidden) rate heterogeneity. While standard GeoSSE models, which are a subset of the Cladogenetic State Speciation and Extinction (ClaSSE) model (Goldberg et al. 2011), link diversification directly to geographic states, they can produce false positives if other unmodeled factors drive speciation. GeoHiSSE addresses this by incorporating hidden states that capture variation in diversification rates not explained by elevation alone, such as microhabitat specialisation or historical climatic shifts (Caetano et al. 2018). This approach ensures a more robust comparison between geography-linked and geography-independent diversification scenarios.

Our analysis involved fitting 10 distinct GeoSSE and GeoHiSSE models (Table 2). These models varied in complexity, testing whether diversification rates are coupled with geographic range or if they vary independently of elevation. We also compared models that treat range contraction as a process distinct from lineage extinction to better understand transitions between highland and lowland habitats. By including models with hidden states, we accounted for potential rate heterogeneity that might otherwise be incorrectly attributed to elevational shifts, providing a more nuanced interpretation of evolutionary dynamics (Caetano et al. 2018).

For our geographical analysis, we assumed uniform taxon sampling. For the 700 m criterion, for lowland areas, we sampled 86% of species (233 out of 271), whereas in highland areas we sampled 78% (70 out of 90). For the alternative elevational thresholds, the proportions varied slightly. Under the 300 m criterion, we sampled 86% of lowland species (208 out of 241) and 79% of highland species (95 out of 120). Under the 1500 m criterion, the corresponding figures were 57% for lowland species (12 out of 21) and 86% for highland species (291 out of 340). We used the function *GeoHiSSE()* in the R package *HiSSE* (Caetano et al. 2018) to fit the models to the phylogeny and range data, reconstruct ancestral ranges under each model, and compare model support using AIC. We used the default setting to fit the 10 different GeoSSE and GeoHiSSE models.

2.5 | Correlation Between Diversification and Environmental Proxies

To estimate tip-level speciation rates for each species, we used the Diversification Rate (DR) metric (Jetz et al. 2012). For this, we used the *DRstat()* function in the R package *epm* (Title et al. 2022). The DR metric considers the number of branching events and the distances between each tip and the root, with greater weight assigned to branches closer to the present (Title and Rabosky 2019). While originally developed as a statistic for estimating diversification (i.e., speciation-extinction) rates, DR has been shown to be more informative for estimating speciation rates alone (Title and Rabosky 2019). Additionally, this method is less computationally intensive than other approaches when applied to large phylogenies (e.g., BAMM; Title and Rabosky 2019). Lastly, to evaluate the relationship between topographic complexity, past climatic-change velocity, and DR, we performed a phylogenetic generalised least squares (PGLS) analysis using the *pgls()* function from the *caper* R package (Orme et al. 2023). To assess model performance, we compared the model fit

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TABLE 2 | GeoHiSSE results. We show the different models we fitted and how they differ in their range effect, separate extirpation, hidden states, and number of parameters. We also show the AIC and the AICc values for each model. The model with the highest AIC weight is shown in bold and italic. Based on Table 1 in Thomson et al. (2021).

Model in *R*	Model	Model description	Range effect	Separate extirpation	Hidden states	Number of parameters	AIC	AICc	AIC weights
1	GeoSSE	No range effect, dispersal varies among areas	No	No	No	4	2825.02	2825.16	8.71E-49
2	GeoSSE	Separate extirpation, dispersal varies among areas	No	Yes	No	6	2800.39	2800.67	1.95E-43
3	GeoHiSSE	No range effect, hidden state effect	No	No	Yes	7	2773.22	2773.60	1.54E-37
4	GeoHiSSE	Separate extirpation, no range effect, hidden state effect	No	Yes	Yes	9	2669.82	2670.44	4.38E-15
5	GeoHiSSE	No range effect, four hidden states effect diversification	No	No	Yes	13	2715.51	2716.77	5.25E-25
6	GeoHiSSE	Separate extirpation, no range effect, four hidden states	No	Yes	Yes	15	2696.66	2698.34	6.50E-21
7	GeoSSE	Range effect (original GeoSSE model)	Yes	No	No	7	2618.31	2618.69	0.00067
8	GeoSSE	Separate extirpation, range effect	Yes	Yes	No	9	2610.99	2611.60	0.026
9	***GeoHiSSE***	***Range and hidden state effects***	***Yes***	***No***	***Yes***	***15***	***2603.75***	***2605.43***	***0.97***
10	GeoHiSSE	Separate extirpation, both range and hidden state effects	Yes	Yes	Yes	19	2621.02	2623.70	0.00017

using the corrected Akaike Information Criterion (AICc; Sugihara 1978; Burnham and Anderson 2002). Additionally, to account for potential clade-specific responses to ecological drivers, we also performed the PGLS analyses separately for clades (e.g., Dactyloa [mostly mainland species], Digilimbus [mostly island species], Ctenonotus [island species], Norops [island species], Draconura [mostly mainland species]), which differ in both phylogenetic history and biogeographic context (e.g., mainland vs. island distribution). In this regard, we can test whether a correlation exists between speciation rates and topographic complexity and past climatic-change velocity, and whether this correlation is clade-specific.

3 | Results

Our final datasets included 233 lowland and 70 highland species based on the 700 m criterion (Figure 1), 208 lowland and 95

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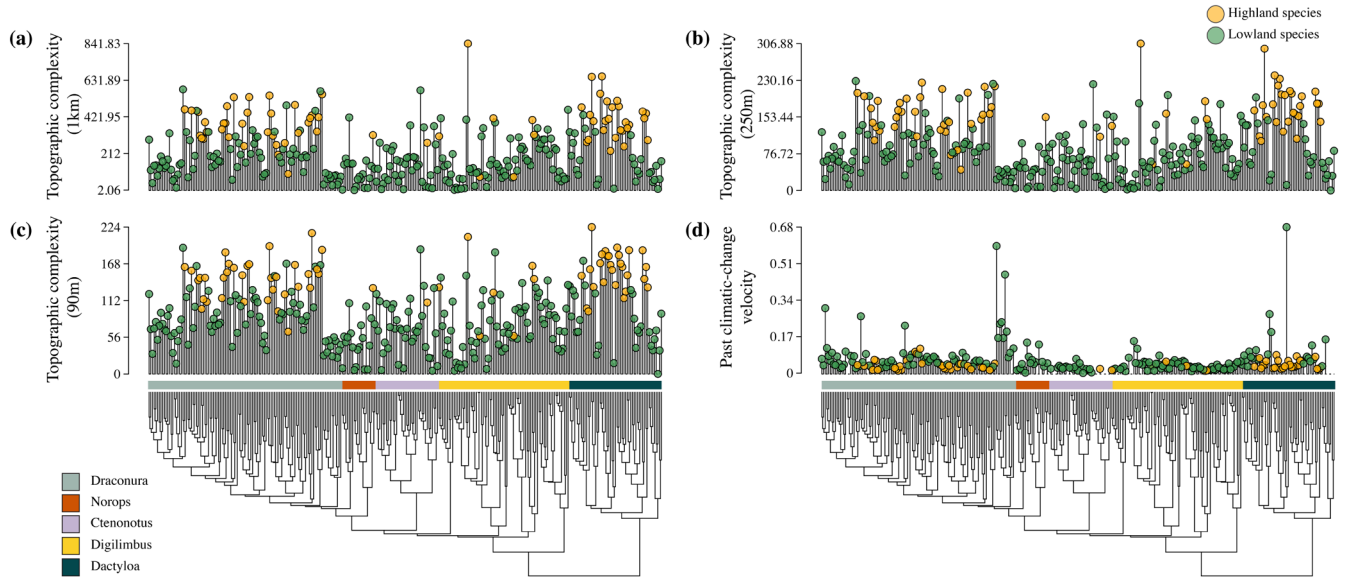


FIGURE 1 | Topographic complexity at three spatial scales 1 km (a), 250 m (b), and 90 m (c), and past climatic-change velocity (d) across the *Anolis* phylogeny. All variables are shown for each tip, representing the mean value for all species assigned to each terminal branch. We indicate whether the species is a highland or lowland species and to which clade each terminal branch belongs.

highland species based on the 300 m criterion, and 291 lowland and 12 highland species based on the 1500 m criterion; the results for the 300 m and the 1500 m criteria can be found in the supplements (Table S2).

3.1 | Temporal Pulses in Neotropical *Anolis* Diversification

We observed that diversification rates declined toward the present (Figure 2a). This decline appears to have begun around the onset of tectonic activity in Central America and continued until the orogeny of the Andes, after which diversification rates stabilised (Figure 2a). Bayes factors (BF) from alternative rate shift models provided positive support (2 ln BF between 2 and 6; Kass and Raftery 1995) for a diversification rate shift occurring during the Miocene, approximately 7 million years ago.

3.2 | Diversification Dynamics Across Highland and Lowland Lineages

Based on the 10 different models we ran for the GeoHiSSE analysis, each criterion (300 m, 700 m, 1500 m) has a different best-supported model; for the case of the 700 m criterion, the best-supported model, GeoHiSSE (AIC weight = 0.97; Figure 2b; Table 2; Table S3), indicated that the ancestral *Anolis* was a widespread species (Figure 2; Figures S1 and S2) with a high net diversification rate (Table 2); we found similar results for the other elevational thresholds (best model, for the 300 m and the 1500 m criteria was GeoSSE; Figures S1 and S2; Table S3). The GeoHiSSE analysis also inferred that most transitions to highland habitats occurred during the late Eocene and the Oligocene (Figure 2b). Additionally, when we compared the speciation rate and net diversification between highland and lowland species, we found that lowland species had higher speciation rates and net diversification compared to their highland counterparts (Figure 2c); this was consistent for the 300 m and the 1500 m criteria.

3.3 | Environmental and Topographic Drivers of Diversification Rates

When we examined the different predictors that might influence speciation rate, we found that topographic complexity was not significantly correlated with speciation rate (1 km resolution, p-value = 0.48, $F_{1,263} = 0.49$; 250 m resolution, p-value = 0.26, $F_{1,263} = 1.28$; 90 m resolution, p-value = 0.19, $F_{1,263} = 1.72$; Table 3; Figure 3a–c; Figure S3). Moreover, there was no significant correlation between speciation rate and past climatic-change velocity (p-value = 0.61, $F_{1,263} = 0.26$; Table 3; Figure 3e; Figure S3). These results suggest that topographic complexity and past climate velocity do not have a strong influence on speciation rate measured using a tip-level metric. When we separated our analysis by clade, we found no significant correlation between speciation rate and topographic complexity or past climatic-change velocity for any of the clades, with the exception of Draconura, for which we found a significant correlation between speciation rate and topographic complexity (1 km resolution p-value = 0.031, $F_{1,109} = 4.77$; 250 m resolution p-value = 0.037, $F_{1,109} = 4.47$; and 90 m resolution p-value = 0.038, $F_{1,109} = 4.39$; Figure 4; Table 3; Figures S4 and S5) and past climatic-change velocity (p-value = 0.010, $F_{1,109} = 6.81$; Figure 4; Table 3; Figures S4 and S5). Statistical results for the other clades may be found in the Supporting Information (Figures S4 and S5; Table S4).

4 | Discussion

Montane regions are known to host most of the hotspots of diversity and endemism in the natural world (Rahbek, Borregaard,

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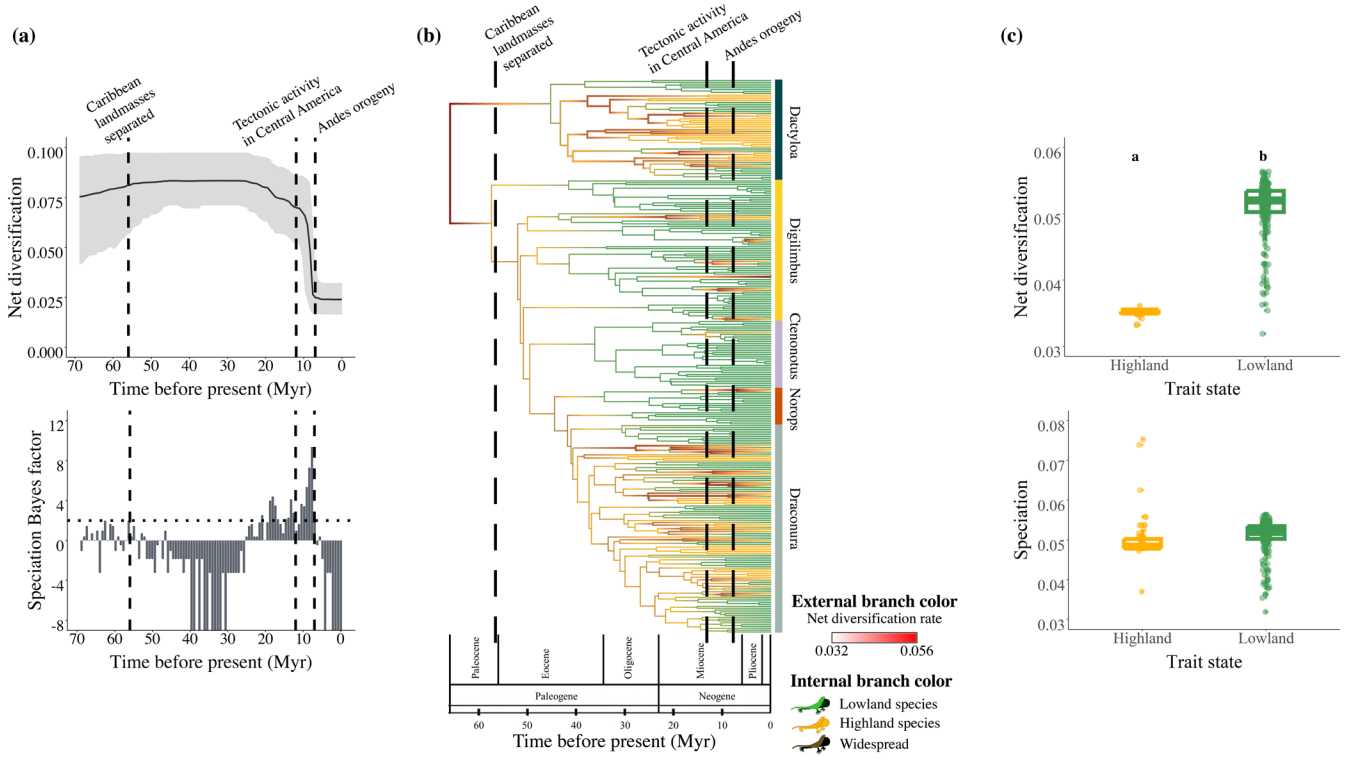


FIGURE 2 | (a; top) Tree-wide diversification rates over time, as inferred from the CoMET analysis. The grey line represents the estimated diversification rate (with the mean and 95% highest posterior density) based on the anole phylogeny shown in (b). (a; bottom) Bayes Factor (BF) evidence for a shift in speciation rates across the tree. The dashed line marks the threshold for "positive" support (2 ln BF > 2; Kass and Raftery 1995). (b) Ancestral range reconstruction of the GeoHiSSE results. The outlines of the branches are coloured by net diversification rates, with white indicating low rates and red indicating high rates. The inner parts of the branches are coloured by reconstructed ancestral states: lowland species (green), highland species (yellow), and widespread species (brown). Different clades of anole lizards are also shown. (c) Estimates of net diversification (top) and speciation (bottom) in highland than lowland areas (for a more in-depth explanation see Data S1).

TABLE 3 | Results from phylogenetic generalised linear regression models for speciation rate (meanDR) assuming Brownian motion as the evolutionary model between the topographic complexity (1 km, 250 m, and 90 m), and past climatic-change velocity. Coefficient, *p*-value, *F*-statistic, degree of freedom (DF), and R^2 values are shown for the *pgls* results. Significant results are shown in italics.

Model	Coefficient	*p*	*F*	DF	R^2
All species					
MeanDR—Topographic complexity (1 km)	0.010	0.68	0.17	1, 263	0.00064
MeanDR—Topographic complexity (250 m)	0.0032	0.81	0.055	1, 263	0.00021
MeanDR—Topographic complexity (90 m)	−0.0028	0.87	0.029	1, 263	0.00011
MeanDR—Past climatic-change velocity	0.017	0.42	0.66	1, 263	0.0025
Draconura					
MeanDR—Topographic complexity (1 km)	*−0.000027*	*0.031*	*4.77*	*1, 109*	*0.040*
MeanDR—Topographic complexity (250 m)	*−0.000066*	*0.037*	*4.47*	*1, 109*	*0.038*
MeanDR—Topographic complexity (90 m)	*−0.000086*	*0.038*	*4.39*	*1, 109*	*0.037*
MeanDR—Past climatic-change velocity	*0.051*	*0.010*	*6.81*	*1, 109*	*0.059*

Note: Coefficient, *p*-value, *F*-statistic, degree of freedom (DF), and R^2 values are shown for the *pgls* results. Significant results are shown in italics.

Colwell, et al. 2019). Previous studies have found that habitat variety associated with the topographic complexity that mountain uplifting creates (Hughes and Eastwood 2009; Hoorn et al. 2010; Rahbek, Borregaard, Colwell, et al. 2019; Rahbek, Borregaard, Antonelli, et al. 2019; Igea and Tanentzap 2021; García-Rodríguez et al. 2025), and climatic stability (measured here as past climatic-change velocity; Dynesius and Jansson 2000; Rolland et al. 2014; Fine 2015; Morales-Barbero

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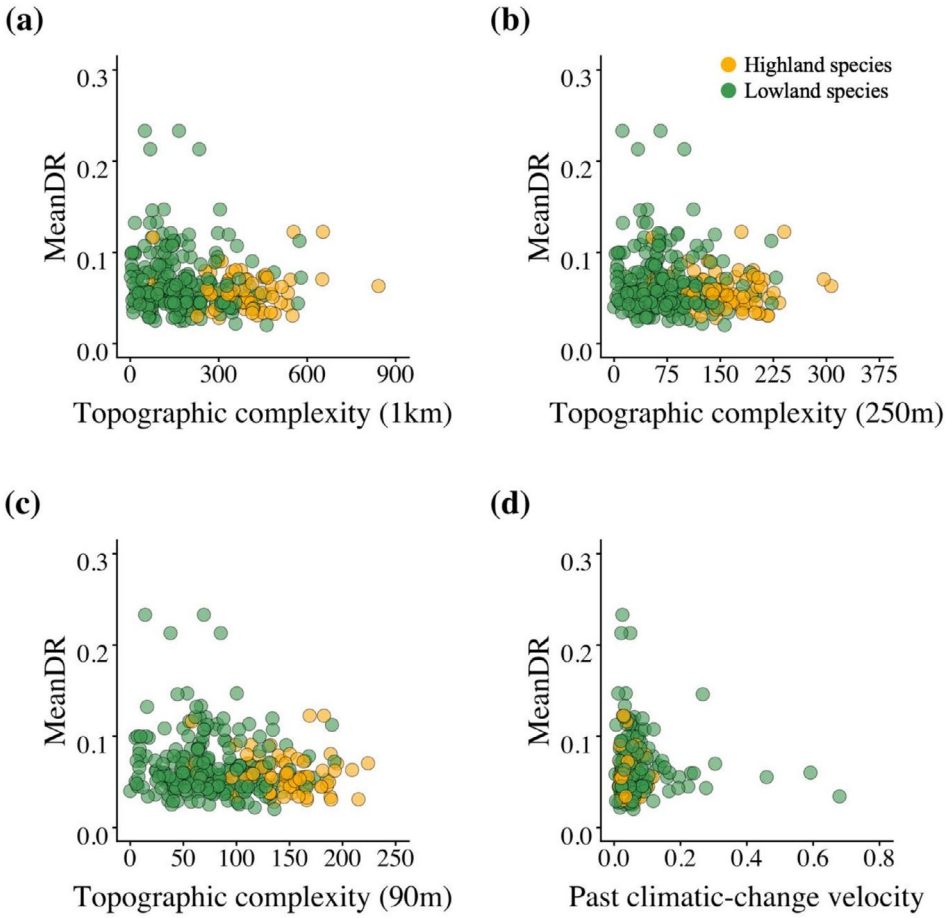


FIGURE 3 | Relationships between mean speciation rate (MeanDR) and environmental variables across the *Anolis* phylogeny using the 700 m criterion. Environmental variables include topographic complexity at three spatial scales 1 km (a), 250 m (b), 90 m (c), and past climatic-change velocity (d). Points represent individual species.

et al. 2021; Igea and Tanentzap 2021; Buckley et al. 2024; García-Rodríguez et al. 2025) is positively related to speciation rates.

Here, we found that lowland species have higher speciation and diversification rates (Figure 2). However, when we tested which factors were responsible, we found that speciation and diversification rates were not related to greater topographic complexity or past climatic-change velocity (Figure 3). In the following sections, we expand on these findings and discuss their implications for understanding diversification processes in montane regions.

4.1 | Biogeographical Context of Diversification

Previous studies have shown that colonisation of mountain regions is often associated with increased speciation rates, especially at high elevations (Moore and Donoghue 2009; Lagomarsino et al. 2016). Our results do not support this pattern for anole lizards: lowland species exhibit significantly higher speciation rates than highland species (Figure 2). However, based on these previous studies, we hypothesized that topographic complexity and climatic stability (measured as past climatic-change velocity) would positively influence speciation rates by promoting long-term isolation and niche specialisation. Specifically, topographic complexity and past climatic-change velocity showed no significant relationship with speciation rate. These findings challenge the generality of previous findings that topographic complexity and past climatic-change velocity are related to higher speciation rates (e.g., Moore and Donoghue 2009; Lagomarsino et al. 2016; García-Rodríguez et al. 2025) and highlight the need to consider alternative pathways to explain the montane speciation patterns.

Other factors that may influence speciation rates in anole lizards (e.g., biotic interactions, habitat connectivity, species-specific dispersal abilities, productivity, time-integrated area, e.g., Losos and Schluter 2000; Buckley and Jetz 2007; Gillman et al. 2009; Igea and Tanentzap 2020; García-Rodríguez et al. 2021, 2025) were not measured in our analysis. However, recent studies (e.g., Wacker and Winger 2024) show that ecological interactions and dispersal limitations can significantly mediate the effects of topography and climate on diversification. For instance, in Neotropical montane birds, speciation rates were more strongly associated with dispersal barriers and ecological interaction intensity than with topographic metrics alone (Wacker and Winger 2024). In our case, some *Anolis* lineages may be more dispersive or ecologically generalist, enabling them to persist across heterogeneous landscapes without undergoing speciation. Conversely, other clades may be more ecologically constrained. Without trait-level or interaction data, it is difficult

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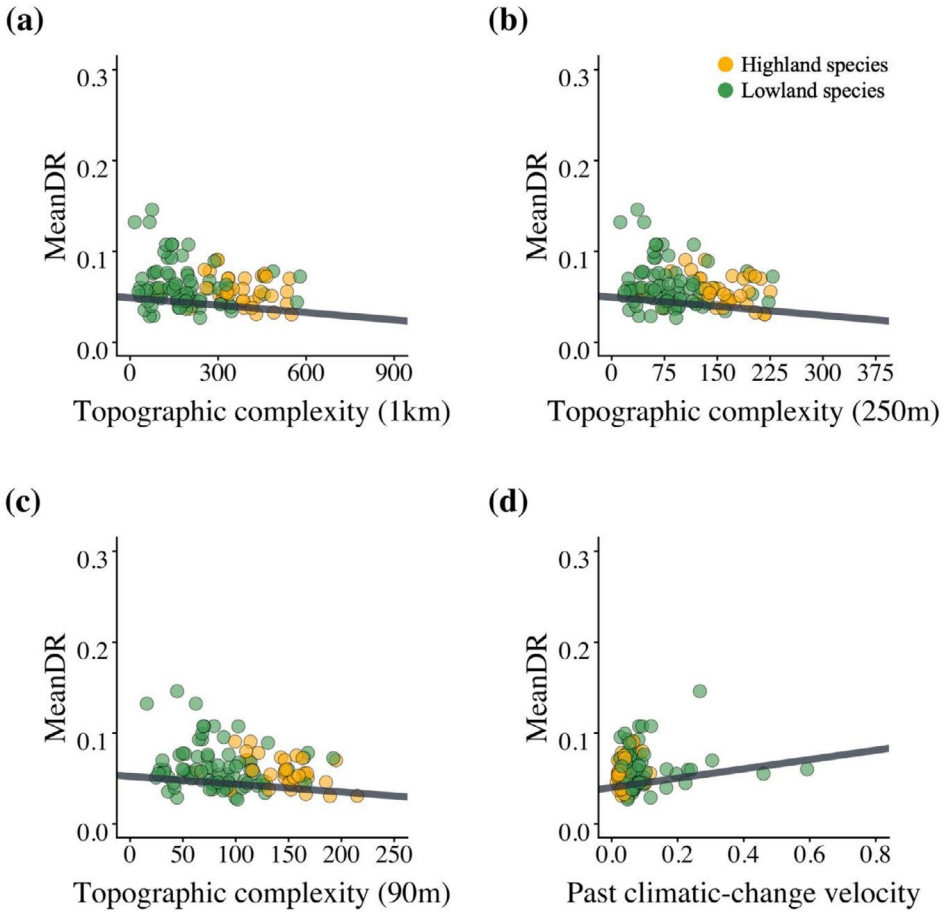


FIGURE 4 | Relationships between mean speciation rate (MeanDR) and environmental variables for Draconura using the 700 m criterion. Environmental variables include topographic complexity at three spatial scales 1 km (a), 250 m (b), 90 m (c), and climatic-change velocity (d). The black regression line represents the results of the phylogenetic generalised least squares (PGLS). Points represent individual species.

to capture these dynamics. Therefore, incorporating ecological traits and measures of interaction strength in future analyses may help explain why certain species diversify in complex environments while others do not.

Nonetheless, for the Draconura clade, the most recent clade of anoles (Poe et al. 2017), we found a different pattern. In this clade, speciation rate decreases with increasing topographic complexity (Figure 4), which suggests that the Draconura clade may possess unique ecological or evolutionary characteristics compared to the other anole clades (Rabosky and Glor 2010). One possibility is that Draconura may have traits that favour broader ecological niches or enhanced dispersal abilities, enabling them to occupy diverse habitats without undergoing speciation (Wiens 2004). Alternatively, increased topographic complexity might impose ecological constraints that limit niche differentiation, thereby reducing the speciation potential within this particular clade (Gavrilets and Losos 2009). While these hypotheses are consistent with observed macroevolutionary patterns, we did not explicitly test for ecological niche breadth, dispersal capacity, or other functional traits due to limited trait data available across species in this clade. However, future comparative work that incorporates trait-based analyses (e.g., range size, habitat specialisation, or dispersal proxies) would be valuable for evaluating these potential mechanisms. These findings underscore the importance of considering clade-specific traits and ecological dynamics when examining patterns of speciation.

4.2 | Timing of Elevational Transitions and Diversification Dynamics

Our state-dependent diversification analysis (GeoHiSSE) revealed that most transitions into highland environments occurred during the late Eocene and Oligocene (~40–25 Mya; Figure 2a). This time frame coincides with a period of increased tectonic activity in the Caribbean and early mountain formation events in Central America and northern South America, suggesting that emerging elevational gradients may have opened new ecological opportunities for colonisation and diversification (Poe et al. 2017).

When comparing diversification patterns across elevational groups, we found that lowland species consistently exhibited higher speciation and net diversification rates than their highland counterparts across all three elevation thresholds (700 m, 300 m, and 1500 m criteria; Figure 2; Figures S1 and S2). These results contrast with findings from other clades where mountain colonisation is associated with high speciation rates (e.g., Moore and Donoghue 2009; Lagomarsino et al. 2016). Nevertheless,

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this difference reinforces a role for elevationally structured habitats in influencing in promoting ecological diversification and diversification, but suggests their effect may be more variable than previously appreciated.

Our temporal diversification analyses further indicated a decline in net diversification rates toward the present (Figure 2). This slowdown appears to have begun during the Miocene (~23–5 Mya) and corresponds with key geological events in the region, including the Andean orogeny and continued uplift in Central America. Notably, a diversification shift was supported by Bayes factor analysis (2 ln BF = 2–6; Figure 2; Kass and Raftery 1995), with the most likely rate change occurring during the Miocene (~7 Mya). These findings suggest a scenario in which diversification accelerated following colonisation of montane habitats but later stabilised as available ecological space filled or dispersal barriers were overcome, a pattern evident in other taxa (vertebrates—Antonelli et al. 2018; birds—Harvey et al. 2020; Igea and Tanentzap 2021; turtles—Thomson et al. 2021). One possible explanation is that the lower average diversification rates observed in the highlands may reflect the fact that the initial burst of highland diversification has already subsided, whereas lowland diversification has remained relatively elevated, potentially due to greater ecological or geographic opportunities.

Although the apparent decline in diversification could reflect ecological saturation, it may also be influenced by unobserved extinction events, especially given the limited fossil record of *Anolis* (Poe et al. 2017). Nevertheless, our phylogenetic reconstructions point to a critical evolutionary window during the Oligocene, Miocene, and Pliocene, during which geological and elevational dynamics likely shaped the modern diversity of this lizard group.

Finally, while our study applied a broad-scale framework that treats all highland regions as ecologically and topographically equivalent, we acknowledge that this approach may obscure important regional differences. For instance, the ecological and geological characteristics of highlands in the Greater and Lesser Antilles may differ from those in mainland Central and South America, potentially influencing diversification in distinct ways (Iturralde-Vinent 2006; Antonelli et al. 2018). This simplification was a necessary trade-off for testing general patterns across broad spatial and phylogenetic scales, but it also highlights the importance of region-specific studies that incorporate localised climate histories, tectonic processes, and lineage-specific traits. Ultimately, refining how highland systems are compared across taxa and space will be key to fully understanding the role of mountains in evolutionary diversification.

4.3 | Past Climatic-Change Velocity and Speciation Rates

Additionally, we found no significant relationship between past climatic-change velocity and speciation rate in Caribbean anoles (Figure 3). This contrasts with previous studies that have reported a positive association between climatic-change velocity and diversification in other vertebrate groups (e.g., Morales-Barbero et al. 2021). In our study, we estimated past climatic-change velocity as the rate of change (km/year) in both temperature and precipitation variables last approx. 3 million years (see Methods). This approach captures a period of rapid climatic change and is consistent with recent studies using paleoclimatic velocities as proxies for climatic stability (e.g., García-Rodríguez et al. 2025). However, given that *Anolis* originated between 46.3 and 64.4 million years ago (Figure 2; Poe et al. 2017), it is possible that the timescale of past climatic-change velocity used here (~3.3 mya) is too recent to meaningfully explain macroevolutionary patterns of speciation across the entire clade. Furthermore, the relationship between climate and diversification may be moderated by other interacting factors, including ecological opportunity, productivity, or biotic interactions, which could obscure direct effects of recent climatic instability. These findings suggest that while paleoclimatic velocity is a valuable proxy for environmental stability, its impact on diversification may vary depending on the evolutionary timescale, geographic context, and clade-specific dynamics under study. For instance, the interplay between species' elevational range constraints and climate could buffer populations from rapid shifts in past climatic-change velocity, reducing its influence on speciation (Sandel et al. 2011; Dullinger et al. 2012; García-Rodríguez et al. 2025). Lastly, integrating different types of data (e.g., fossil records, ecological data, geospatial data, mountains orogeny time) with a broader evolutionary timescale could provide a more informative view of how speciation rates are influenced by factors other than topographic complexity and past climatic-change velocity.

When we analysed clades separately, the Draconura clade exhibited distinct patterns compared to the others. Specifically, we found that the speciation rate within this clade increased with increasing past climatic-change velocity (Figure 4), suggesting that rapid environmental changes may promote faster diversification in these lizards. This result suggests that regions with higher past climatic-change velocity have acted as a boost for the speciation rate the Draconura clade experienced over the past 3 million years. However, it is important to note that this clade, being the youngest, has undergone numerous speciation events recently (Poe et al. 2017). These recent events could potentially bias our results. To better understand these patterns, future research should focus on refining methods to assess the influence of past climatic changes over long evolutionary timescales (Fenton et al. 2023).

4.4 | Future Perspectives

Future studies should focus on establishing an elevation threshold that accounts for regional variation in climate, topography, and species distributions. Such a threshold would better reflect the ecological and evolutionary processes that shape species' adaptation to elevation and would allow for comparisons across different montane regions. Understanding how these varying delimitations influence our estimates of mountain colonisation and evolution remains a key priority, especially in the context of climate change. As climate change alters temperature and precipitation patterns, it is crucial to understand how these shifts impact species distributions and evolutionary trajectories in montane regions. Further studies should also focus on exploring whether diversification rates align or precede specific biogeographical and topographic shifts. A holistic approach, one that incorporates both climatic and topographic complexity, is essential for predicting species

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responses to future climate changes. This approach would provide insights into speciation and adaptation mechanisms across varied landscapes, especially those with high climatic homogeneity and high topographic heterogeneity. Understanding these dynamics can help build conservation strategies to protect biodiversity amid the impacts of global climate change.

Author Contributions

Jhan C. Salazar developed the original idea, analysed the data, and contributed to the manuscript draft. Adam C. Algar contributed to the manuscript draft. Steven Poe gathered the data and contributed to the manuscript draft. Jonathan B. Losos developed the original idea and contributed to the manuscript draft. Julián A. Velasco gathered the data, developed the original idea, analysed the data, and contributed to the manuscript draft. All authors contributed to the discussion of the results and provided substantively to the manuscript development.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors confirm that the data supporting the findings of this study are available within the article Supporting Information—S1. And codes used for analysis and figures can be found in FigShare: https://figshare.com/s/d3716e463b7656d01359.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** geb70218-sup-0001-FigureS1-S5-TableS1-S4.docx. **Data S2:** geb70218-sup-0002-TableS2-S2.csv.

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